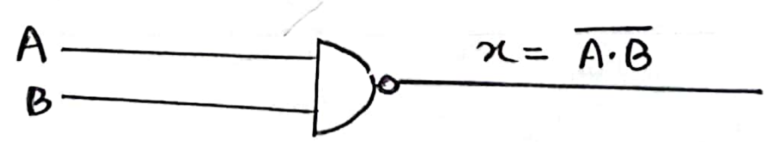
Name of the experiment : Verify the universality of NAND gate and NOR gate.

Objective: The objective of this experiment is NAND and NOR gates can be used to implement any Boolean function.

Theory : Universality of NAND gate :

All Boolean expression of the basic operation of NAND, OR, NOT. Therefore, any expression can be implemented by using only NAND gate and no other types of gates. This can be used to perform each of the Boolean expression.

Symbole : NAND gate :



Truth Table :

Input		Output
A	B	$x = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

NAND gate replace to NOT gate :

A NAND gate with its inputs connected together behaves as an 'inverter'. That's means NAND gate acts as not gate.

The output is

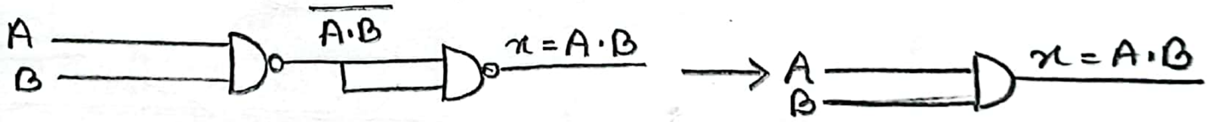
$$x_0 = \overline{A \cdot A}$$
$$= \overline{A}$$

Symbole :

NAND gate replace to AND gate :

Two NAND gates are arranged to perform the AND operation. Two NAND gate2 is used as an 'inverter' to change $\overline{A \cdot B}$ to $\overline{\overline{A \cdot B}}$ which is the desired AND function.

Symbole :



Truth table :

Input		Output
A	B	$x = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

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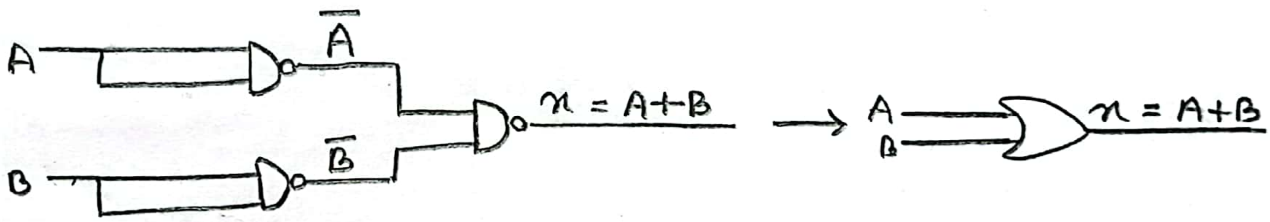
NAND gate replace to OR gate :

OR operation can also be implemented by using 3 NAND gates. Here gates 1 and 2 are as inverters to invert the inputs. So that, the final output is $x = \overline{\bar{A} \cdot \bar{A}}$ which can be simplified to $x = A+B$ using Dr morgan Law.

Symbole :



Truth Table :

Input		Output
A	B	$x = \overline{\bar{A} \cdot \bar{B}} = A+B$
0	0	0
0	1	1
1	0	1
1	1	1

Input output
2, 3 = 1
5, 6 = 4
8, 9 = 10
11, 12 = 13

Universility of NOR gate is not given in our question

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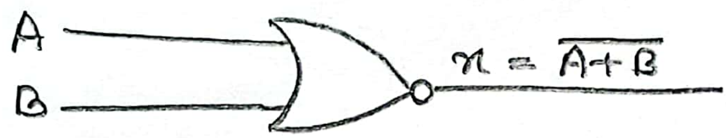
Universality of NOR gate:

All Boolean expressions consists of various combination of the basic operation of AND, OR, NOT. Therefore, any expression can be implemented by using only NOR gate. This can be perform each of the Boolean expressions.

Symbole:

NOR -gate:



Truth Table:

Input		Output
A	B	$x = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

NOR gate replace to NOT gate:

A NOR gate with its inputs connected together behaves as an inverter that mean a NOR gate acts as a NOT gate. The output is

$$x = \overline{A+B} = \overline{A}$$

Symbole:

Truth Table:

Input	Output
A	$y = \overline{A+A} = \overline{A}$
0	1
1	0

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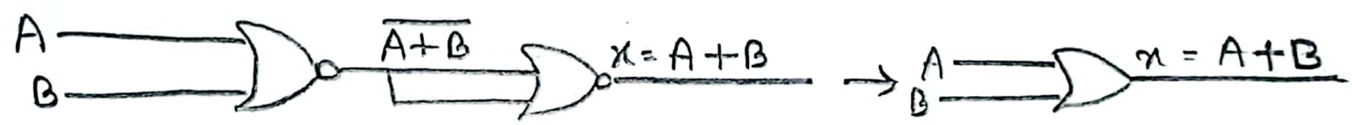
NOR gate replace to OR gate :

Two NOR gates are arranged so that the OR operation is performed. NOR gate +2 is used as an inverter to change $\overline{A+B}$ to $\overline{\overline{A+B}} = A+B$ which is designed OR function.

Symbole :



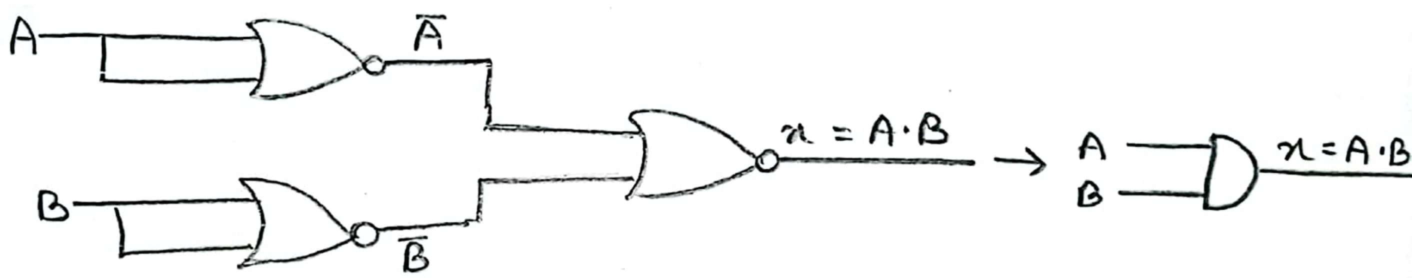
Truth Table :

Input		Output
A	B	$Y = \overline{\overline{A+B}} = A+B$
0	0	0
0	1	1
1	0	1
1	1	1

NOR gate replace to AND gate :

AND operation can also be implemented by using 3 NOR gates. Here gate 1 and 2 are used as inverters to invert the inputs. So that the final output is $x = \overline{\overline{A}+\overline{B}}$ which be simplified to $x = A \cdot B$ using D'morgan law.

Symbole :



Truth Table :

Input		Output
A	B	$Y = \overline{\overline{A}+\overline{B}} = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

Apparatus :

* NAND gate
* NOR gate
* Bread Board
* Resistor
* LED
* Power supply
* Wire .

Procedure :

At first we have to identify the pin number of an IC . The IC set on the Bread Boad for NAND and NOR gate . Pin 1 and 2 are input and 3 is output . So the wire attached to the input pin . Another wire from the final output pin connected to the resistor and then to the output light . We also connected the Vcc and GND with IC .

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Result :

For NAND gate

A	$Y = \overline{A \cdot A} = \overline{A}$	LED
0	1	ON
1	0	off

A	B	$Y = \overline{\overline{A \cdot B}} = A \cdot B$	LED
0	0	0	off
0	1	0	off
1	0	0	off
1	1	1	ON

A	B	$\overline{A}$	$\overline{B}$	$Y = \overline{\overline{A} \cdot \overline{B}} = A + B$	LED
0	0	1	1	0	off
0	1	1	0	1	ON
1	0	0	1	1	ON
1	1	0	0	1	ON

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For NOR gate

A	$Y = \overline{A+A} = \overline{A}$	LED
0	1	ON
1	0	OFF

A	B	$Y = \overline{\overline{A+B}} = A+B$	LED
0	0	0	OFF
0	1	1	ON
1	0	1	ON
1	1	1	ON

A	B	$\overline{A}$	$\overline{B}$	$Y = \overline{\overline{A}+\overline{B}} = A \cdot B$	LED
0	0	1	1	0	OFF
0	1	1	0	0	OFF
1	0	0	1	0	OFF
1	1	0	0	1	ON

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Discussion :

In this experiment, we have cheacked all the gates of IC's and found the active gates. We found out the position here the IC got proper supply and use these for this experiment.

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